

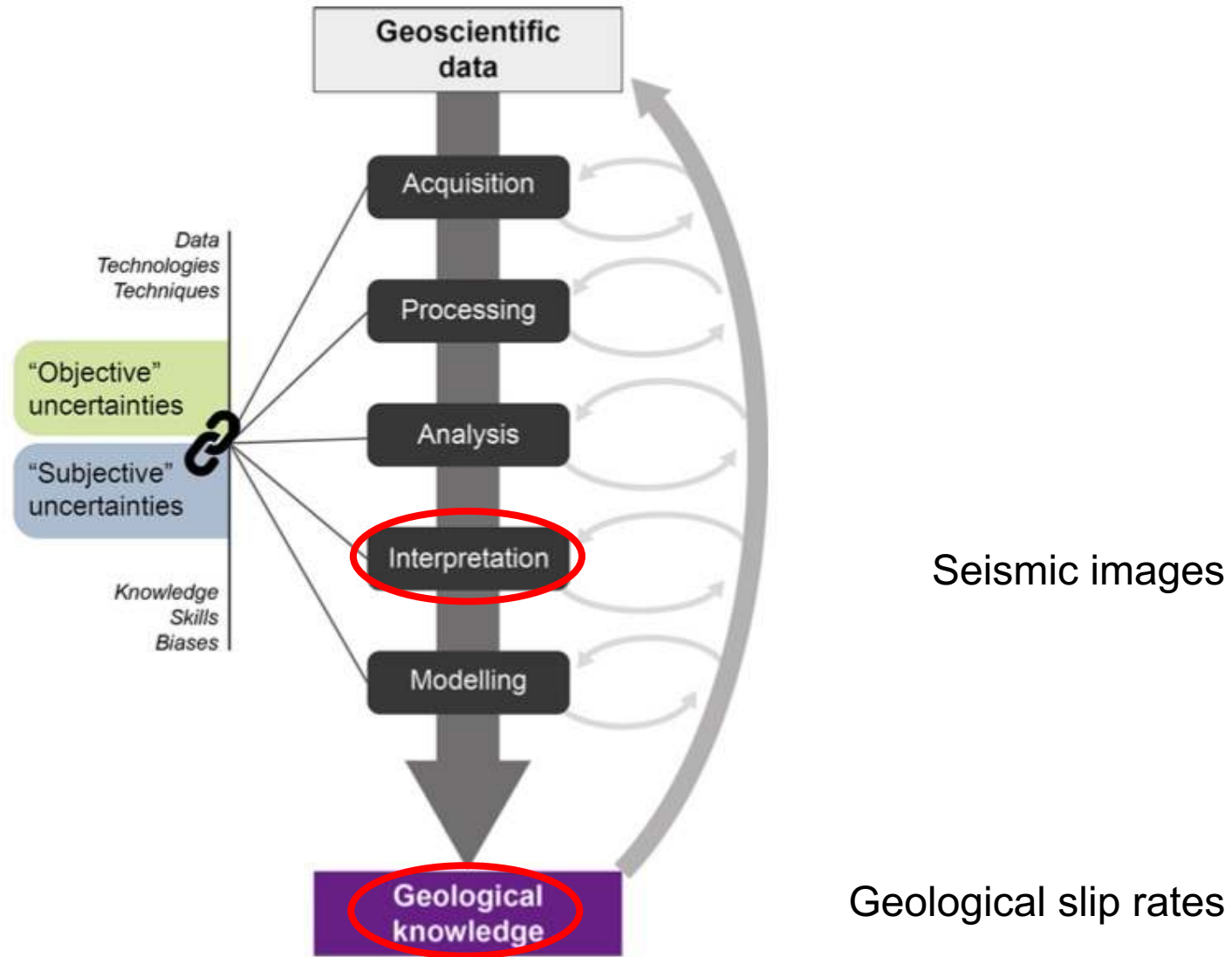
“Is uncertainty useful?”

How do differences in interpreting seismic images affect estimates of geological slip rates?

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Uncertainty throughout all procedures

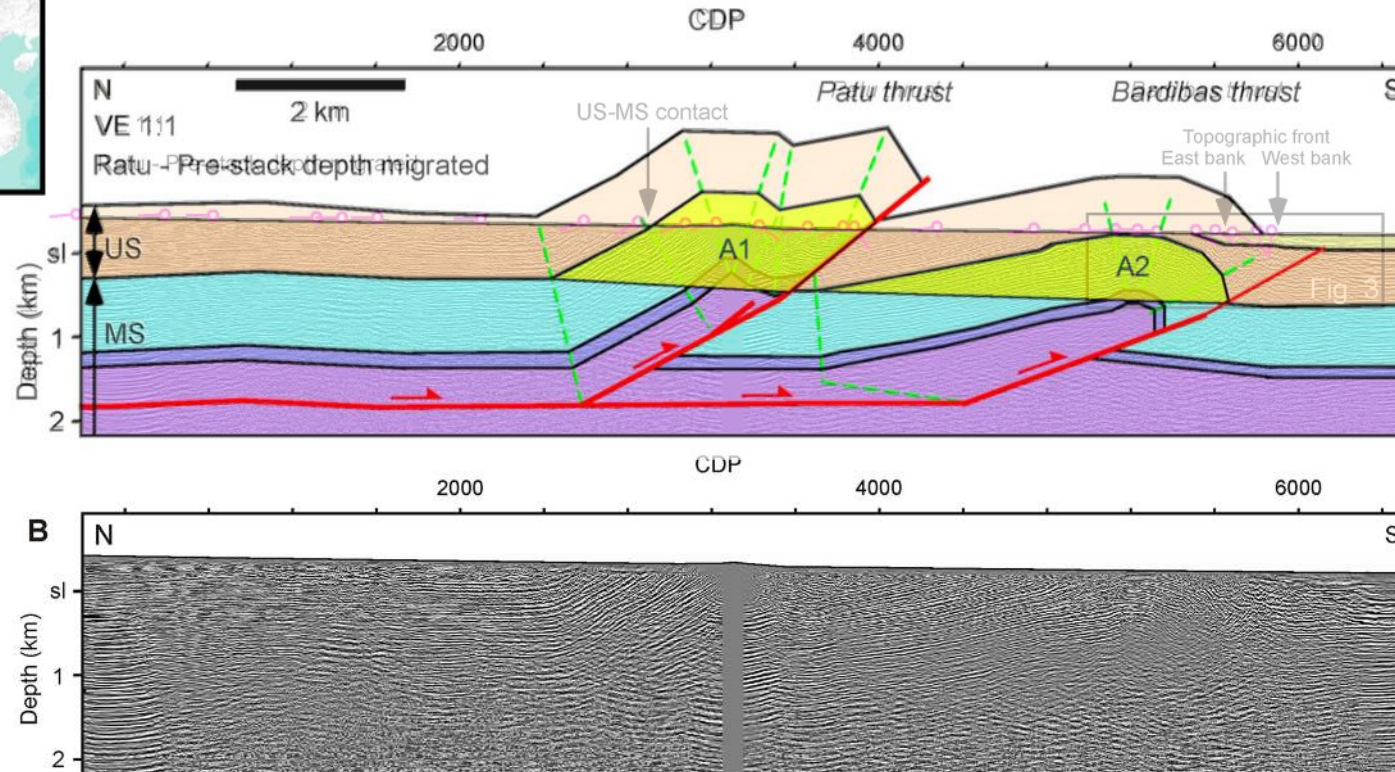
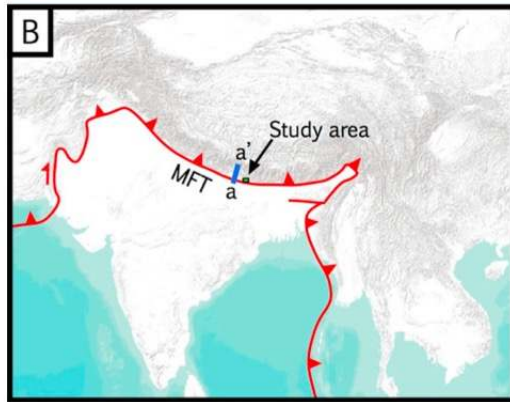


Why interpretation matters

Minor differences in interpreting seismic images could lead to significant differences in estimating geological slip rates (~ 10 mm/year).

From seismic images to geological slip rates

Almeida et al. (2018)



$A2 = 2.21 \text{ km}^2$

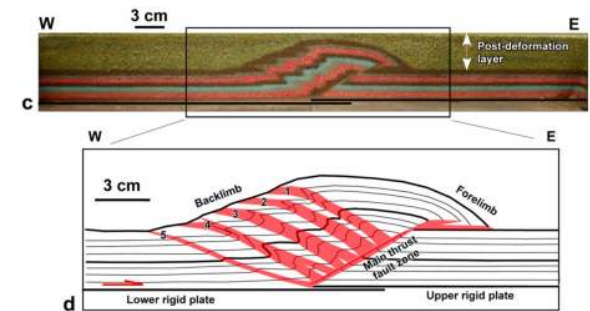
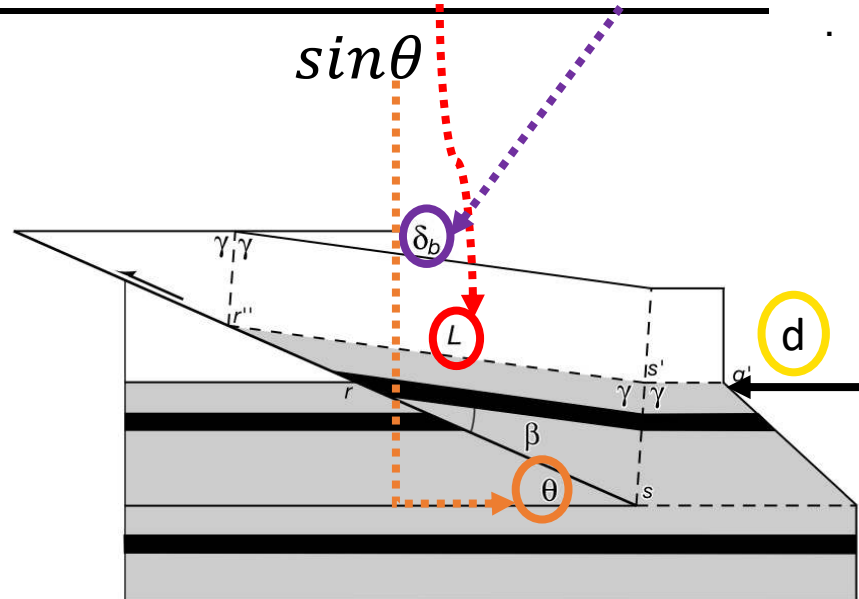
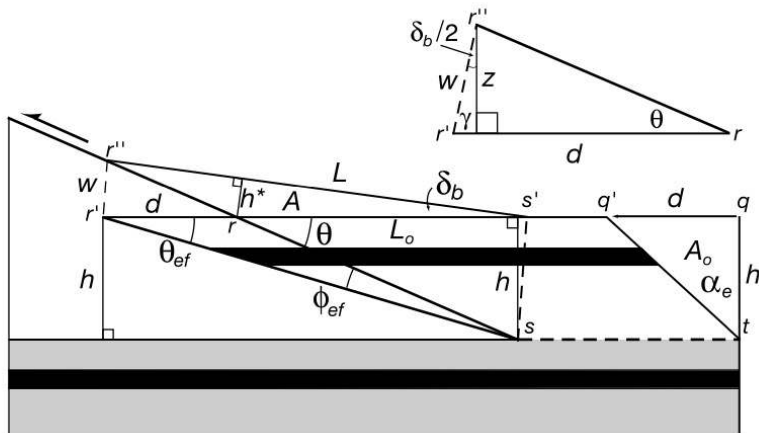
Depth to detachment (Bardibas) = 1.32 km

Shortening (Bardibas) = $\frac{2.21 \text{ km}^2}{1.32 \text{ km}} = 1.67 \text{ km}$

- Get the total shortening, assuming area conservatism
- Get age $\rightarrow$ geological slip rate

A deterministic model for shear fault-bend folding

$$d = \frac{\cos(\delta_b/2 - \theta) \times 2L \sin(\delta_b/2)}{\sin \theta}$$



(Suppe et al., 2004)

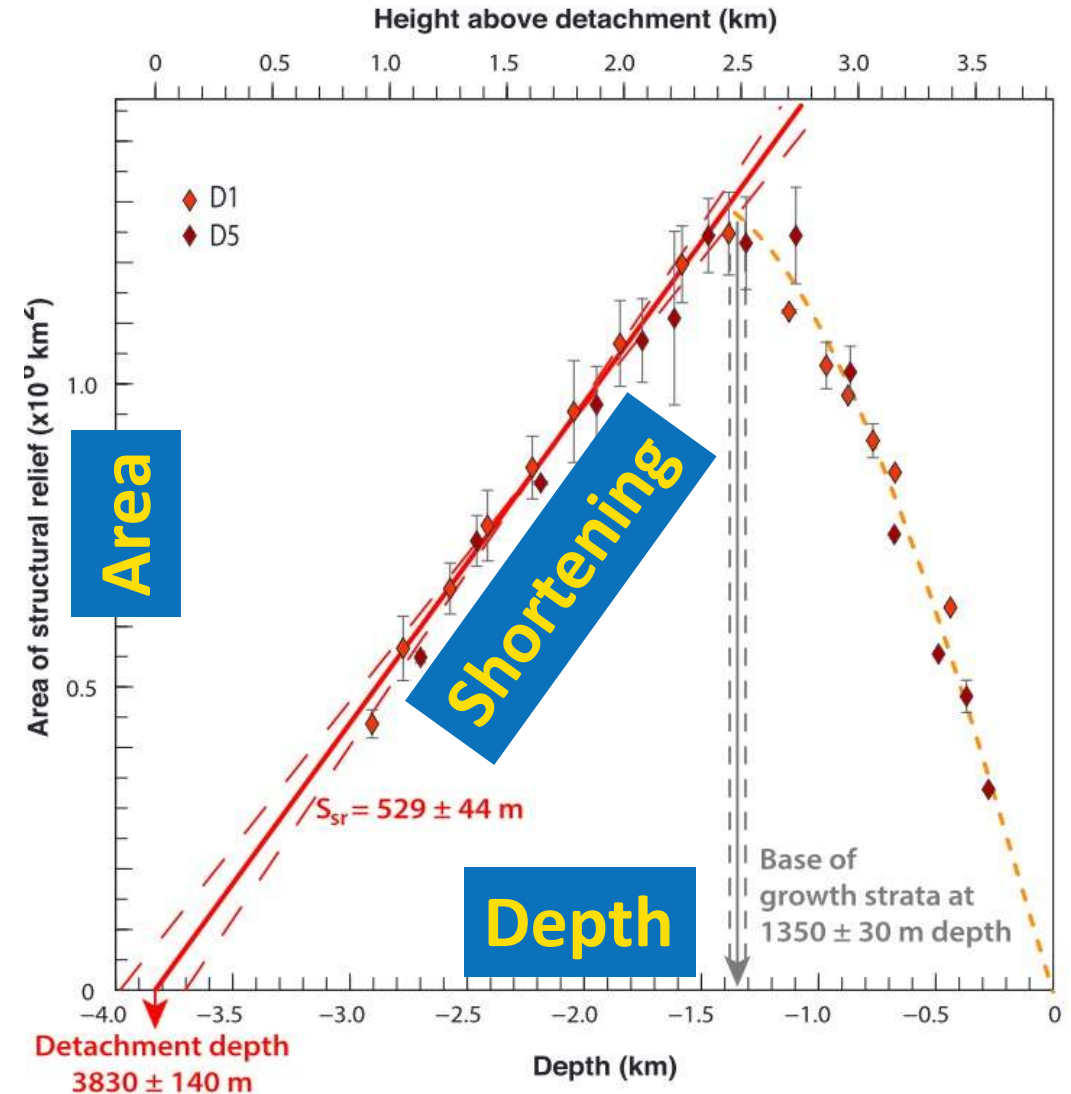
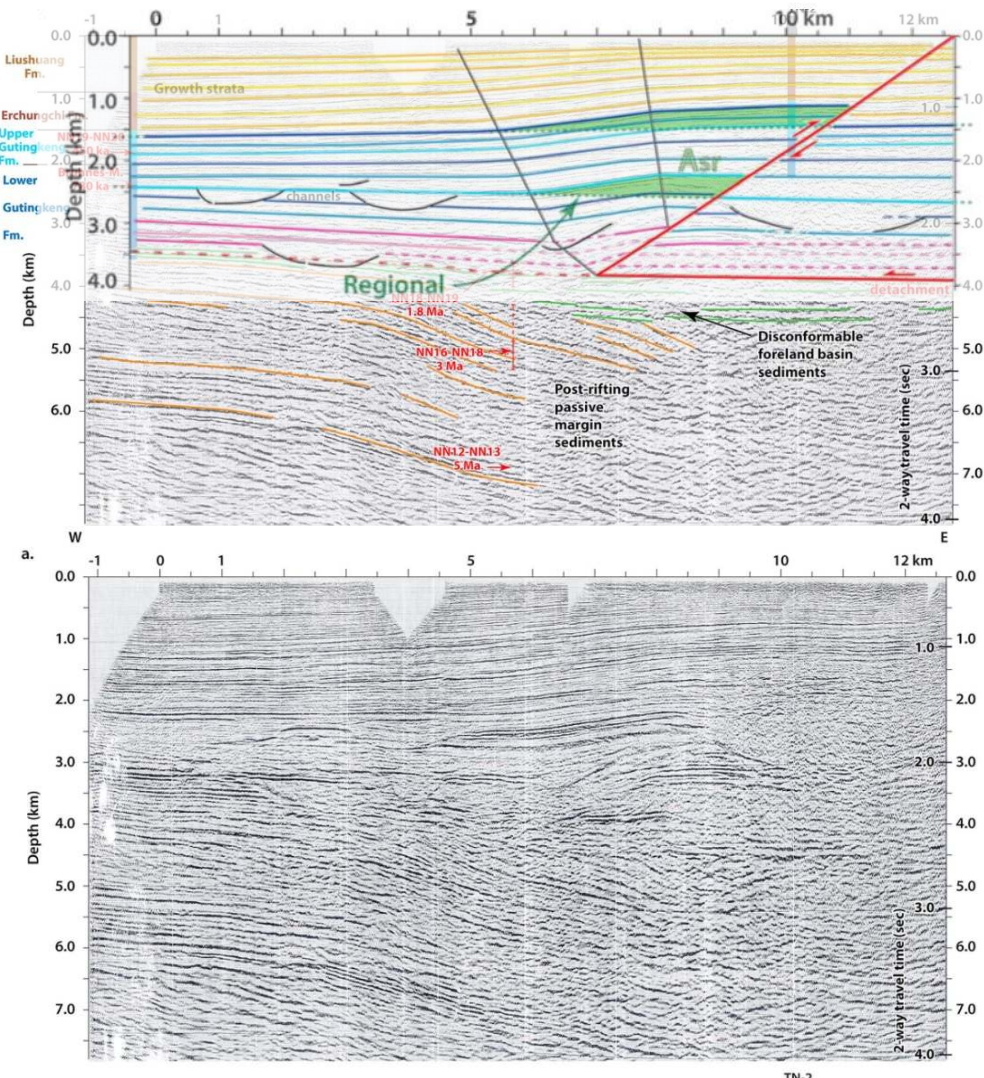
(Groshong et al., 2012)

Test the proposed deterministic model for shear fault-bend folding

The comparable case in Taiwan,
Haochiali backthrust by Le Béon et al. (2019)

Le Béon et al. (2019)

Using Area-Depth-Strain method (Groshong, 2015)

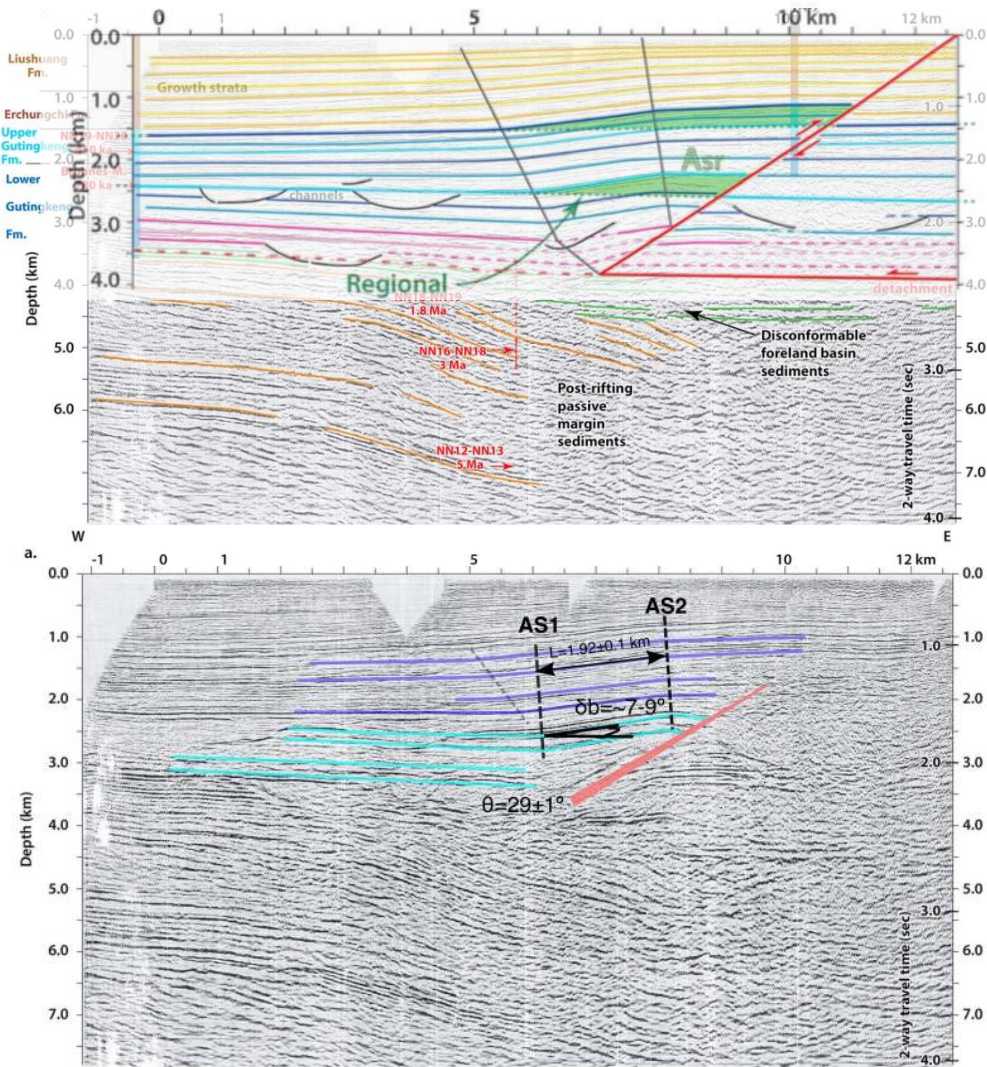


Test the proposed deterministic model for shear fault-bend folding

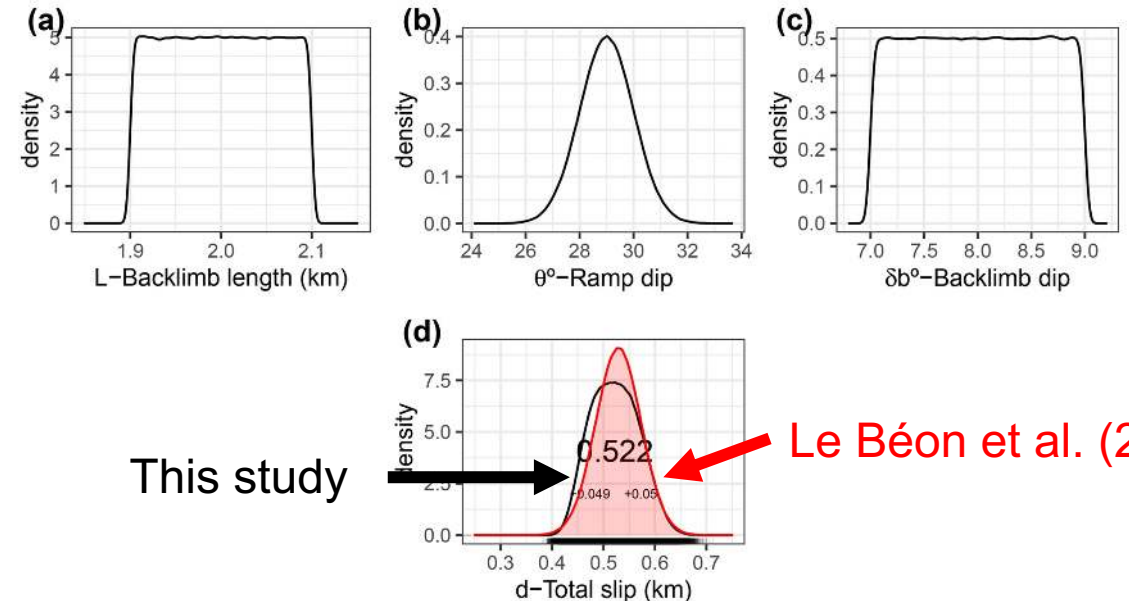
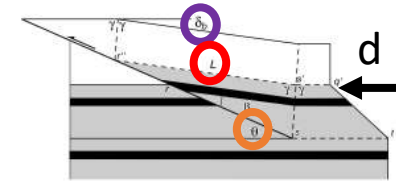
The comparable case in Taiwan,
Haochiali backthrust

This study

Using shear fault-bend folding model (Suppe et al., 2004)
Decide three variables with uncertainty ranges



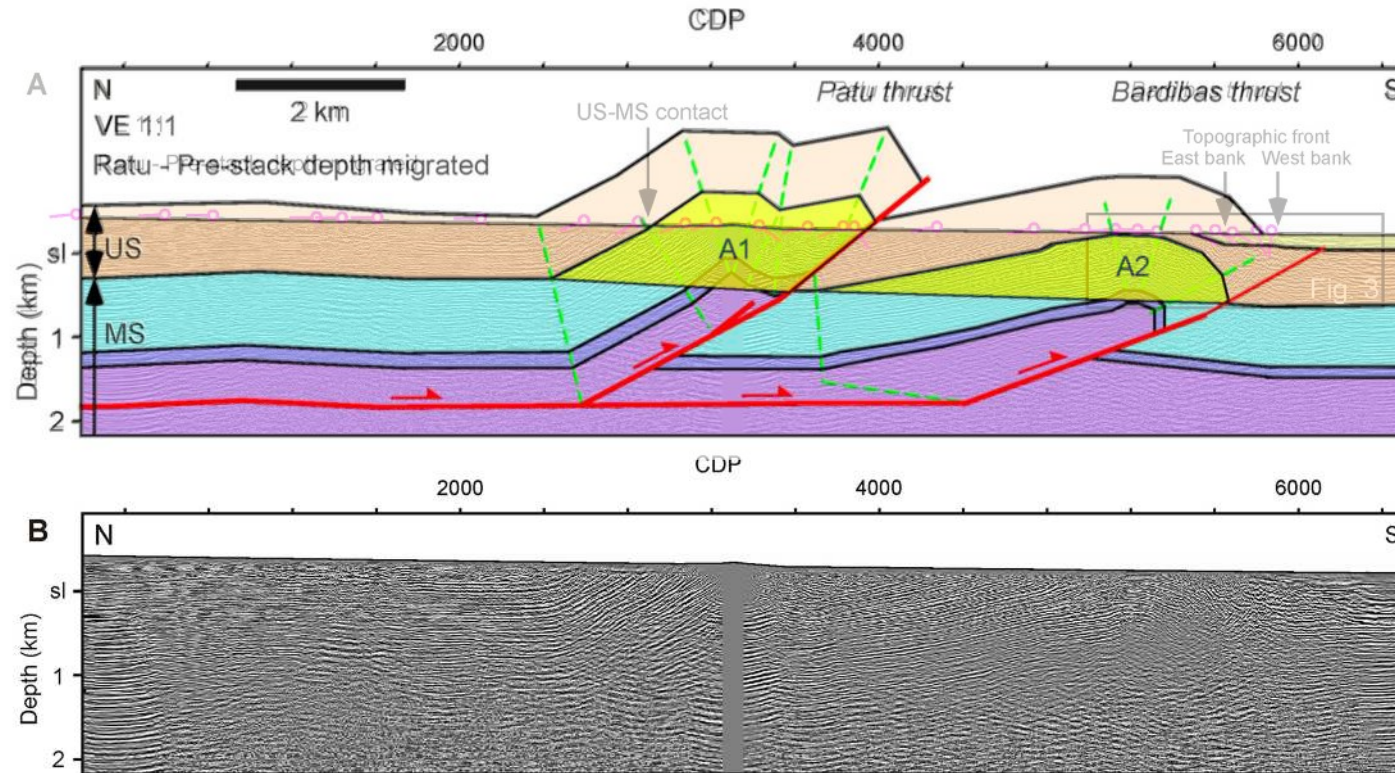
$$d = \frac{\cos(\delta_b/2 - \theta) \times 2L \sin(\delta_b/2)}{\sin\theta}$$



This study

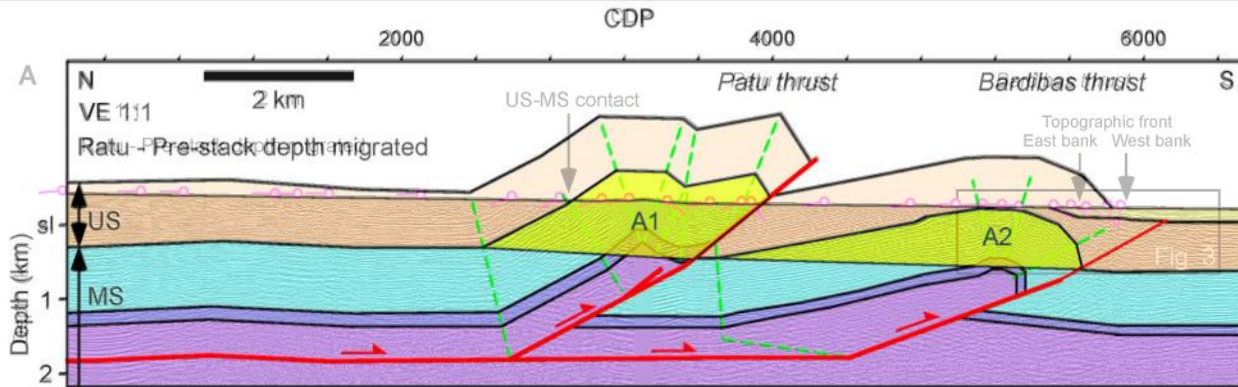
Le Béon et al. (2019)

Apply the deterministic model for shear fault-bend folding to an unclear seismic image

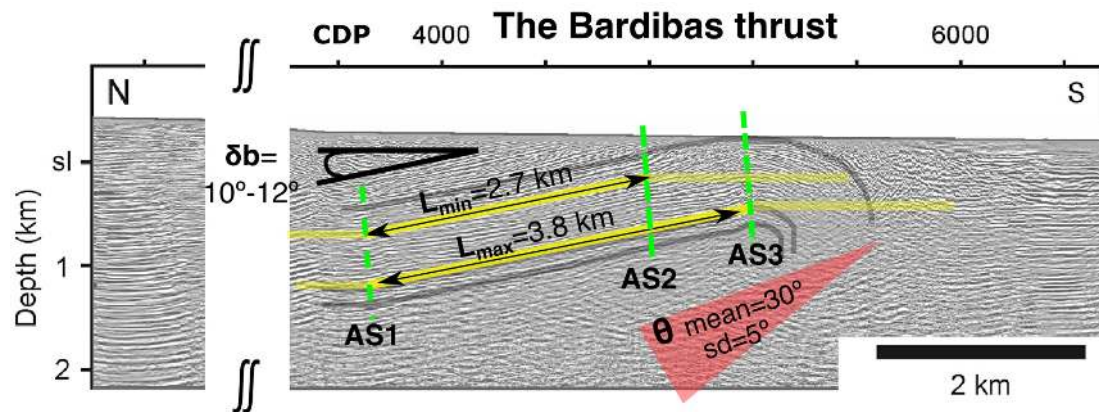
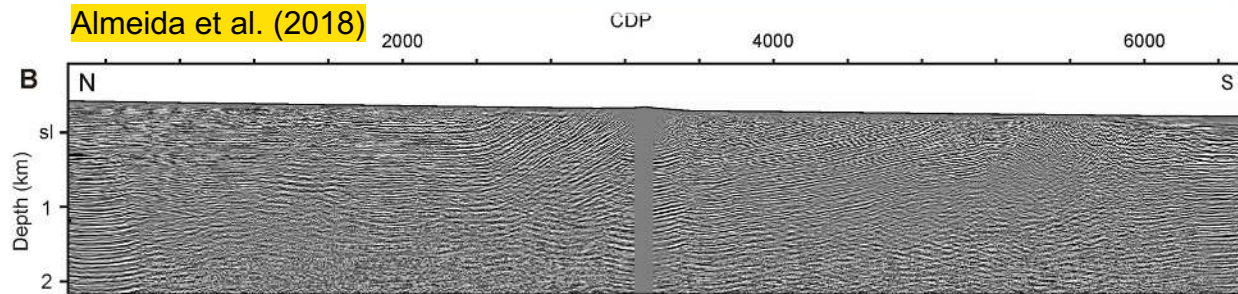


$A2 = 2.21 \text{ km}^2$
 Depth to detachment (Bardibas) = 1.32 km
 Shortening (Bardibas) = $\frac{2.21 \text{ km}^2}{1.32 \text{ km}} = 1.67 \text{ km}$

Compare results of the two studies



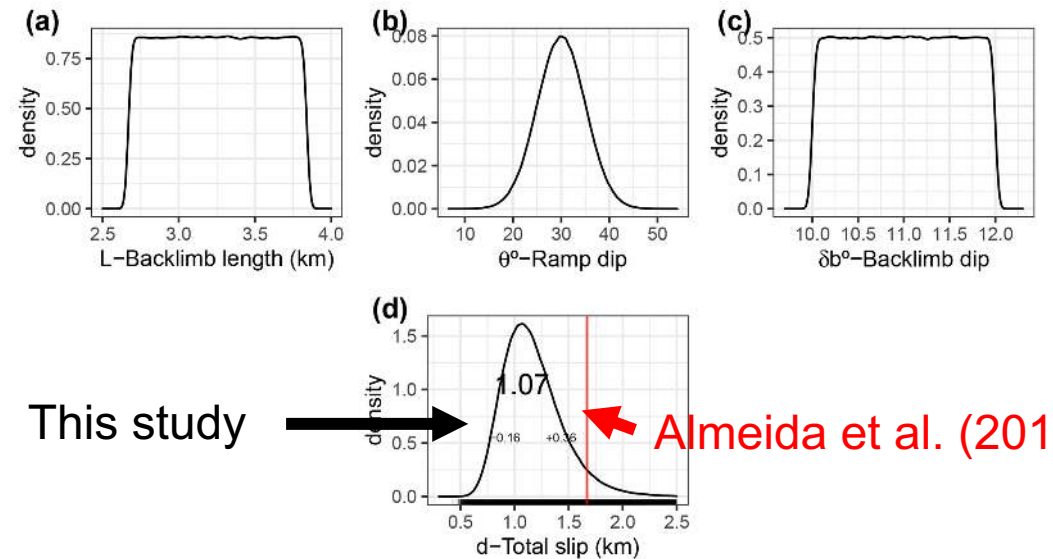
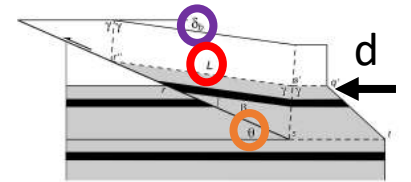
Almeida et al. (2018)



This study

Using shear fault-bend folding model (Suppe et al., 2004)
Decide three variables with uncertainty ranges

$$d = \frac{\cos(\delta_b/2 - \theta) \times 2L \sin(\delta_b/2)}{\sin\theta}$$

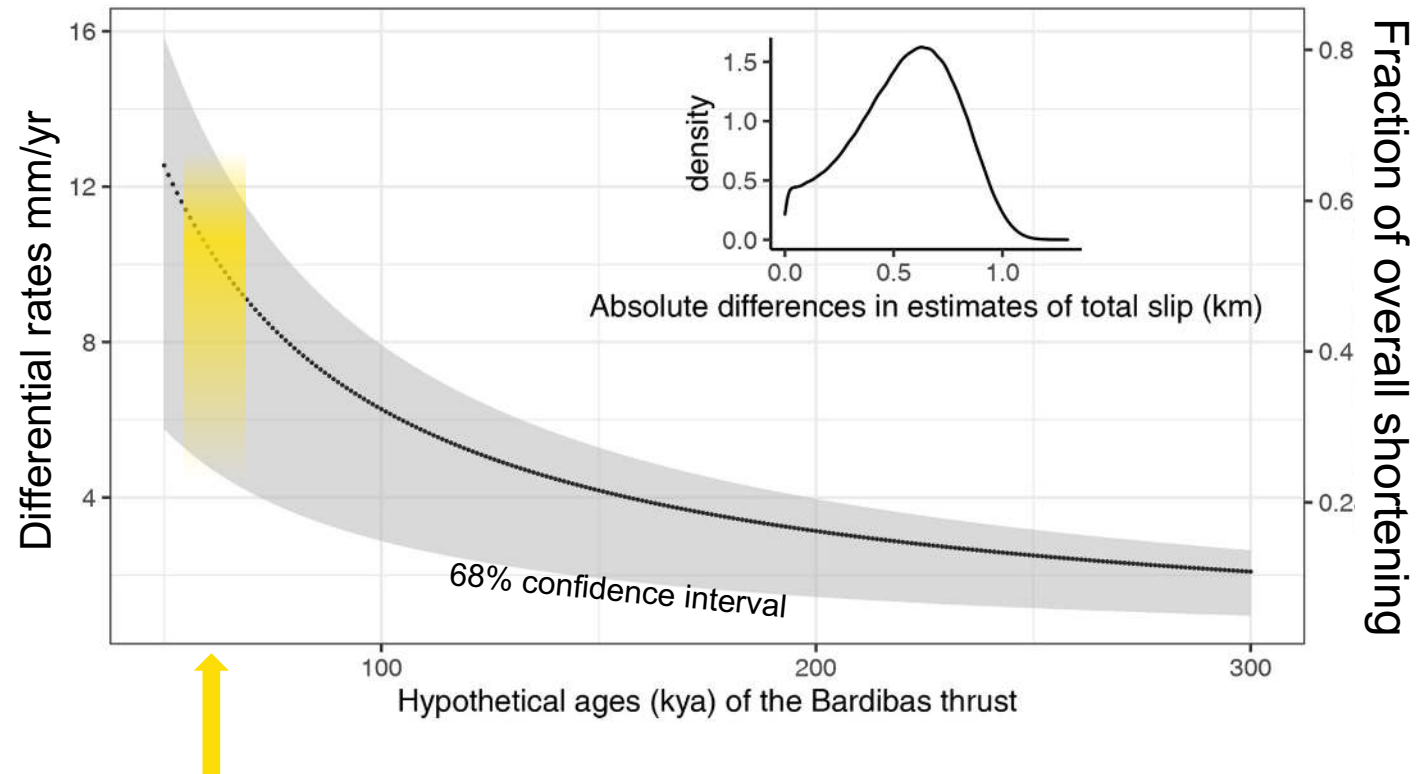


This study

Almeida et al. (2019)

The resulting differential geological slip rates

Differential rates originating from the different estimates of total slip



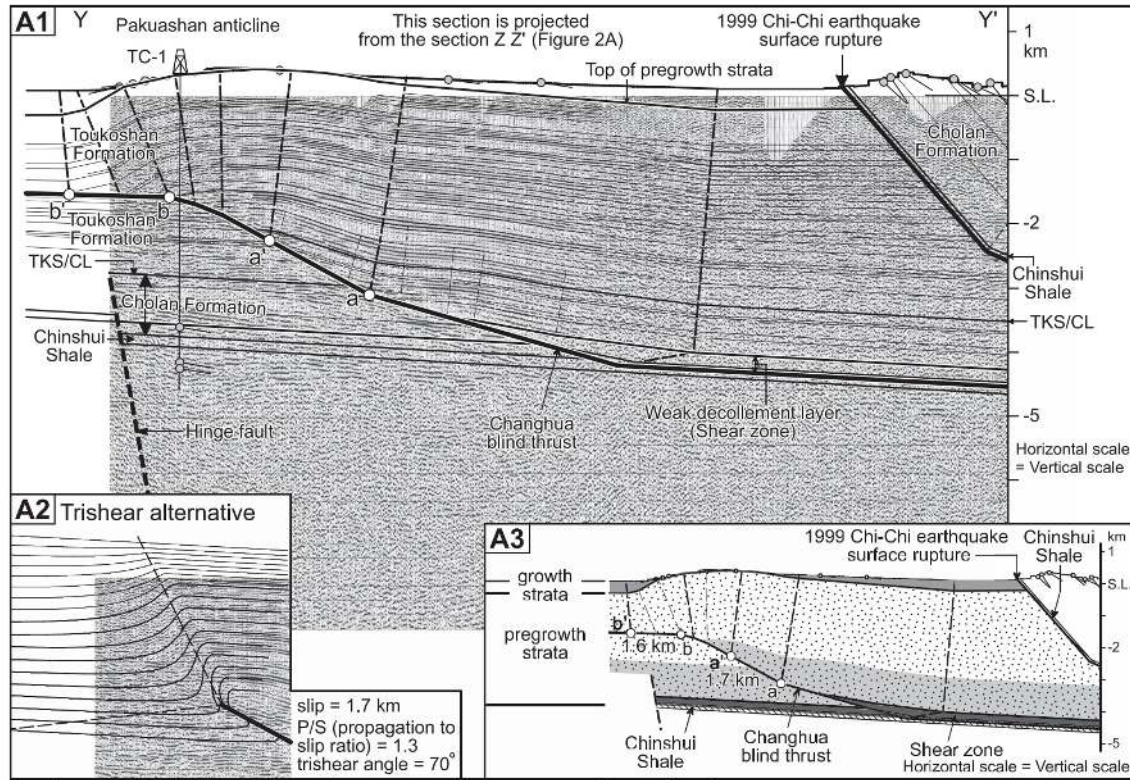
The activated age is ~ 61+ ka, according to geodetic rate ~19.4 mm/yr

Another case in Taiwan (S Pakuashan anticline)

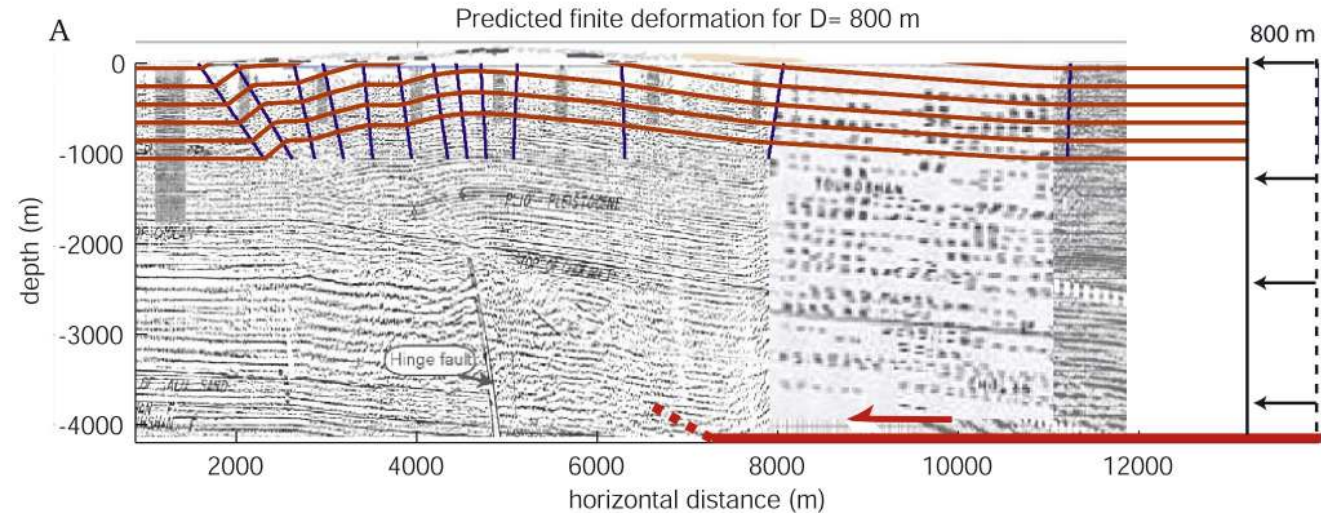
~5 mm/yr

21 ± 5.8 mm/yr

16.3 ± 4.1 mm/yr



(Yue et al., 2011)



Variable estimates of the cumulative shortening across the Pakuashan anticline:

~500 m [Mouthereau et al., 1999],

700 m [Yue et al., 2005],

~4 km [Delcaillau, 2001].

(Simoes et al., 2007)

Take-home message / Why interpretation matters

- Minor differences in interpreting seismic images could lead to significant (~10 mm/yr) differences in estimating geological slip rates.

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